Experiment 6 Day 3: Conversion of Camphene to Camphor

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Introduction

A variety of reagents can convert alcohols into ketones through oxidation. In this experiment, borneol was reacted with sodium hypochlorite, sodium bicarbonate, and 5% sodium bisulfite. With the processes of reflux, liquid-liquid extraction, and the use of drying agents, camphor was synthesized.

Materials and Methods

A long-neck, round-bottom flask was filled with 0.3 mL of racemic borneol product created in the prior lab and 0.3 mL of glacial acetic acid. The mixture was then dissolved in 1 mL of acetone. A reflux distillation apparatus was assembled according to Figure 1-18 on page 20 of the CHEM 2081 Lab Manual, a boiling chip was added to the flask, and the mixture was refluxed in a hot water bath for 35 minutes (Halterman & Morvant, 2013). Over the course of the reflux, 6 mL of sodium hypochlorite was added in small increments and the flask was swirled after each addition. After all of the sodium hypochlorite was added, the mixture was refluxed for 15 more minutes. When the reflux was complete, the mixture was left to cool to room temperature. The camphor mixture was then extracted twice with 2 mL portions of methylene chloride. After this, the organic layer was neutralized with 2 mL of saturated sodium bicarbonate, washed with 2 mL of 5% sodium bisulfate followed by 2 mL of water, and dried with anhydrous sodium sulfate. The product was poured on a watch glass and set over a hot water bath to allow the methylene chloride to evaporate. The weight of the product was noted, all waste was disposed of as instructed, and all used glassware was cleaned under the fume hood with soap and water.

Results

The yield of camphor was 127%.

Calculations

Weight of Watch Glass (g)	38.704
Weight of Watch Glass and Sample (g)	38.974
Weight of Sample (g)	1.27

Table 1: Masses of watch glass, watch glass and sample, and sample alone

$$\frac{1.01\ g\ borneol}{1\ mL\ borneol} x \frac{1\ mol\ borneol}{154.25\ g\ borneol} x \frac{1\ mol\ camphor}{1\ mol\ borneol} x \frac{152.23\ g\ camphor}{1\ mol\ camphor} = 0.977\ g\ camphor$$

$$(\frac{1.27\ g}{0.997\ g})100\% = 127\%\ yield\ of\ camphor$$

Discussion

In this experiment, borneol was oxidized to camphor with the addition of sodium hypochlorite. The oxygen of borneol attacked the sodium hypochlorite, leaving chlorine attached to the alcohol. Once the sodium bicarbonate and sodium bisulfite were added, the chlorine was removed and the carbonyl group was formed, regarding the ketone. Refluxing the solution expedited the exothermic reaction. The process of liquid-liquid extraction allowed for the organic layer to be easily removed from the aqueous layer due to the differences in density between camphor and methylene chloride. The anhydrous sodium sulfate was used to dry any remaining water in the sample. Some sources of error include not evaporating all of the methylene chloride, adding too much sodium hypochlorite solution at one time, and not refluxing the solution for long enough. The source of error that most likely occurred is not fully evaporating the methylene chloride, seeing as though the final yield of the sample was well over 100%. This could have been avoided by allowing the sample to sit over the hot water bath for a longer amount of time.

References

Halterman, R.L., & Morvant, M.C. (2013). *CHEM 2081 Organic Chemistry II Laboratory Manual* [lab manual]. Auburn (Al): Auburn University.